

Highlights

- A four-sport screen preceded locked evaluation in graded cycle ergometry.
- CycHRR-T is bounded, monotonic, and preserves the maximal endpoint.
- Temporal VO₂-reserve MAE improved by 1.07 percentage points versus raw HRR.
- Exact temporal 10% VO₂-reserve-band agreement increased from 50.9% to 59.7%.
- Nonlinear benefit beyond fitted linear calibration was not generally established.